IEEE Student Branch Data Observatory PROJECT

Project Report
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1. Introduction

This project analyzes IEEE Student Branch event data using Python. A dataset of 30 IEEE events was created and analyzed to understand attendance trends, registrations, member retention, event popularity, and budget utilization.

2. Objectives

- Analyze registrations and attendance.
- Identify the most popular event type.
- Study member retention.
- Analyze average budget by event type.
- Create visualizations and a simple dashboard.

3. Tools and Technologies

Programming Language: Python
Libraries: Pandas, Matplotlib
Platform: Google Colab
Version Control: Git & GitHub
Dataset Format: CSV

4. Methodology

The dataset was created and stored in a Pandas DataFrame. Data validation was performed to ensure attendance never exceeded registrations, budgets were positive, and there were no missing values. Statistical analysis was performed using describe(), mean(), sum(), and standard deviation. Visualizations were generated using Matplotlib.

5. Results and Observations

• Hackathons recorded the highest average attendance.
• Workshops had the highest average budget allocation.

• Returning members were higher than new members, indicating good member retention.
• Registration counts were generally higher than attendance.
• A dashboard was created to summarize total events, attendance, registrations, feedback, and budget.

6. Graphs

Insert the following graphs here:
1. Attendance Trends
2. Event Popularity
3. Member Retention
4. Budget Analysis

7. Conclusion

The project demonstrates how Python can be used for data analysis and visualization. The insights obtained from the dataset can help IEEE Student Branch organizers make informed decisions regarding event planning, budget allocation, and member engagement.